HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and deployed a **Python**-based model inference service using **FastAPI** on **AWS Lambda**, processing approximately **500 requests per minute** for real-time scoring.
- Engineered end-to-end **ML pipelines** in **Python** for data preprocessing, feature extraction, and model evaluation, reducing manual data preparation time by **40%**.
- Trained and evaluated a **gradient-boosted classifier** (scikit-learn) on a dataset of **~150,000 customer transactions** to detect anomalous patterns, achieving an **AUC of 0.88**.
- Designed and implemented **containerized ML workflows** using **Docker** and deployed them to an **AWS EKS cluster**, enabling scalable and reproducible experimentation.
- Integrated **LangChain** and a **vector database** (Pinecone) to build a **Retrieval-Augmented Generation (RAG) system**, improving response accuracy for LLM-powered queries by **30%**.
- Contributed to the development of **LLM evaluation frameworks**, writing Python scripts to automate the comparison of model outputs against human-generated judgments for summarization tasks.
- Managed the CI/CD process for ML model deployment using **Jenkins** and Git, ensuring consistent and automated rollouts to production environments with **automated testing**.

SKILLS

Languages – Python, SQL, Java

ML/AI Frameworks – scikit-learn, LangChain, FastAPI, PyTorch, TensorFlow

Data & ML Tools – pandas, NumPy, vector databases, MLflow, ML Ops

Cloud & DevOps – AWS (Lambda, EKS, S3, EC2), Docker, Kubernetes, Jenkins, Git

Databases – PostgreSQL, MySQL, Oracle

Core Concepts – LLM, RAG, Model Training, Inference, Data Pipelines, ETL, CI/CD

PROJECTS

LLM-Powered Customer Support Assistant (MSc Capstone)

- Developed a **RAG system** integrating customer knowledge base documents with an **LLM** via **LangChain**, enabling intelligent query responses.
- Engineered data ingestion pipelines in **Python** to process and chunk documentation for indexing into a **vector database** (Faiss), handling over **5,000 documents**.
- Built a web interface using **Flask** to showcase the assistant's capabilities, allowing users to test its accuracy and provide feedback on response quality.

Production ML Pipeline for Anomaly Detection (HCL Technologies)

- Constructed a scalable **Python** data pipeline on **AWS** to extract features from transaction logs for an anomaly detection model, processing **~2,000 records hourly**.
- Implemented model retraining and evaluation loops, tracking performance metrics like **Precision@K** and **Recall** over time to ensure continued accuracy.
- Containerized the pipeline components using **Docker** and deployed them to an **AWS EC2** instance managed via **Jenkins CI/CD** for automated updates.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)